5.5

> Fixed-cutoff moving-cutoff

1. statement

printed page 1104 equation (5.5) $t < t_0$

$$\rho_{X_0,t_0}(F,t) = \frac{1}{4\pi(t_0-t)}\exp\left(-\frac{|F-X_0|^2}{4(t_0-t)}\right),$$

fixed original-coordinate cutoff $\phi \in C_c^\infty(B_{2r}(X_0))$ $\phi = 1$ on $B_r(X_0)$

$$\Psi(X_0,t_0;t) = \int_{\Sigma_t} (\cos\alpha)^{-p}\phi(F)\rho_{X_0,t_0}(F,t)\,d\mu_t.$$

Theorem 3.3 relevant estimate

$$\frac{d}{dt}\left(e^{c_1\sqrt{t_0-t}}\Psi(X_0,t_0;t)\right) \le -\text{nonnegative square terms} + c_2 e^{c_1\sqrt{t_0-t}}$$

right side one-sided positive error Section 5 first singular point (X_0,T)

$$F_\lambda(x,s) = \lambda(F(x,T+\lambda^{-2}s) - X_0), \qquad d\mu_s^\lambda = \lambda^2 d\mu_{T+\lambda^{-2}s},$$

$w_\lambda = (\cos\alpha_\lambda)^{-p}$ printed equation (5.5) literal endpoint quantity

$$G_\lambda(s) = e^{c_1\sqrt{T-(T+\lambda^{-2}s)}}\int_{\Sigma_s^\lambda} w_\lambda\,\phi_R(F_\lambda)\,(-s)^{-1}e^{-|F_\lambda|^2/[4(-s)]}\,d\mu_s^\lambda,$$

ϕ_R is a compact cutoff fixed in scaled coordinates fixed $-1 < s_1 < s_2 < 0$

$$G_\lambda(s_2) - G_\lambda(s_1) \to 0 \qquad (\lambda\to\infty).$$

continuation remaining moving-cutoff lemma full proof literal compact-scaled-cutoff lemma full Section 5 compact first-singular-flow context full-hypothesis counterexample fixed-original-cutoff correction

2.

status: **proven** Fixed quantity terminal limit $Q : [a,T) \to [0,\infty)$ locally absolutely continuous

$$Q'(t) \le c_2 e^{c_1\sqrt{T-t}}$$

for almost every $t < T$

$$A(t) = \int_t^T c_2 e^{c_1\sqrt{T-u}}\,du, \qquad M(t) = Q(t) + A(t).$$

$A(t) \to 0$ as $t \uparrow T$

$$M'(t) = Q'(t) - c_2 e^{c_1\sqrt{T-t}} \le 0.$$

M monotone decreasing bounded below by 0 $M(t)$ finite one-sided limit $Q(t) = M(t) - A(t)$ finite one-sided limit $t_\lambda(s) = T + \lambda^{-2}s$ fixed $s_1, s_2 < 0$

$$Q(t_\lambda(s_2)) - Q(t_\lambda(s_1)) \to 0.$$

one-sided inequality positive error term

status: **proven** fixed-cutoff rescaled endpoint theorem original-coordinate cutoff $\chi \in C_c^\infty(B_{2r}(X_0))$ $\chi = 1$ on $B_r(X_0)$

$$Q_\chi(t) = e^{c_1\sqrt{T-t}}\Psi_\chi(X_0, T; t).$$

scaled variables expanding cutoff

$$\chi^\lambda(Y) = \chi(X_0 + \lambda^{-1}Y).$$

$$G_\lambda^\chi(s) = e^{c_1\lambda^{-1}\sqrt{-s}} \int_{\Sigma_s^\lambda} w_\lambda\, \chi(X_0 + \lambda^{-1}F_\lambda)(-s)^{-1} e^{-|F_\lambda|^2/[4(-s)]}\, d\mu_s^\lambda$$

exact identity

$$G_\lambda^\chi(s) = 4\pi Q_\chi(T + \lambda^{-2}s).$$

proof sketch

$$T - t = \lambda^{-2}(-s), \qquad F - X_0 = \lambda^{-1}F_\lambda, \qquad d\mu_s^\lambda = \lambda^2 d\mu_t.$$

Kähler angle positive homothety invariant numerator $\omega(u, v)$ area denominator λ^2 $w_\lambda = w$ unnormalized scaled Gaussian factor $4\pi\rho_{X_0,T}$ 4π common normalization factor normalized kernel $[4\pi(-s)]^{-1}$ $Q_\chi(t)$ has a common terminal limit epsilon proof

$$G_\lambda^\chi(s_2) - G_\lambda^\chi(s_1) \to 0.$$

status: **proven** printed cutoff fixed cutoff literal expression

$$\phi_R(F_\lambda) = \phi_R(\lambda(F - X_0))$$

original coordinates moving cutoff

$$\chi_{R,\lambda}(X) = \phi_R(\lambda(X - X_0)).$$

supported in $B_{2R/\lambda}(X_0)$ equal to 1 on $B_{R/\lambda}(X_0)$ λ fixed terminal-limit argument fixed Q_χ moving family $Q_{R,\lambda}$ Q_λ nonnegative has terminal limit $Q'_\lambda \le 0$

$$Q_\lambda(T + \lambda^{-2}s_2) - Q_\lambda(T + \lambda^{-2}s_1) \not\to 0.$$

individual terminal limits for a moving family are insufficient uniform bridge

status: **proven** standalone literal moving-cutoff lemma flat symplectic plane $P = \mathbb{R}^2 \times \{0\} \subset \mathbb{C}^2$ stationary flow $F(x,t) = x$ $\cos\alpha = 1$ $H = 0$ curvature/flow-energy terms vanish $\Sigma_s^\lambda = P$ nontrivial radial compact cutoff ϕ_R

$$I(s) = \int_P \phi_R(Y)(-s)^{-1}e^{-|Y|^2/[4(-s)]}\,dA(Y).$$

full uncut integral 4π compact-cut integral constant $a = -s$ change variables $Y = \sqrt{a}\,Z$

$$I(-a) = \int_P \phi_R(\sqrt{a}\,Z)e^{-|Z|^2/4}\,dA(Z).$$

$s_1 < s_2 < 0$ $a_1 > a_2$ radial nonincreasing cutoff transition annulus scales

$$I(s_2) > I(s_1).$$

$$G_\lambda(s) = e^{c_1\lambda^{-1}\sqrt{-s}}I(s)$$

positive endpoint defect plane full Section 5 counterexample noncompact finite first singular time X_0 singular point

status: **proven / conditional** actual endpoint weighted Radon measures s_1, s_2 weakly converge nonzero two-dimensional cone measure literal compact-scaled-cutoff endpoint difference positive limit compact continuous test function

$$Y \mapsto \phi_R(Y)(-s_i)^{-1}e^{-|Y|^2/[4(-s_i)]},$$

weak convergence integrals two-homogeneous cone radial scaling compact cutoff s-dependence conditional obstruction full counterexample same-limit conical convergence circularly later tangent-cone theorem

3.

wall moving cutoff order-one annular flux

$$\chi_{R,\lambda}(X)=\phi_R(\lambda(X-X_0))$$

fixed-cutoff calculation $D\chi_{R,\lambda}$ scales like λ $D^2\chi_{R,\lambda}$ scales like λ^2 parabolic time window $t=T+\lambda^{-2}s$ $\rho_t=\lambda^2\rho_s$ $d\mu_t=\lambda^{-2}d\mu_s^\lambda$ $dt=\lambda^{-2}ds$ powers cancel

$$\int_{s_1}^{s_2}\int_{\Sigma_s^\lambda} w_\lambda\rho_s\Big[D\phi_R(F_\lambda)\cdot v_\lambda+\Delta_{\Sigma_s^\lambda}(\phi_R\circ F_\lambda)+2\langle\nabla\log\rho_s,\nabla(\phi_R\circ F_\lambda)\rangle\Big]\,d\mu_s^\lambda ds.$$

integral supported in fixed scaled annulus $R<|F_\lambda|<2R$ small λ factor sign local area bounds fixed-time compactness zero velocity/zero curvature heuristics stationary plane pure spatial heat-kernel flux already equals the nonzero endpoint defect later time-independent tangent-cone convergence pre-(5.5) proof (5.5) estimates

4. Approach timeline

stage	question addressed	conclusion established	effect on the approach
fixed terminal quantity	Does one-sided estimate imply a finite terminal limit?	yes, by adding the integrable future error tail	validates the common-limit argument for one fixed cutoff
exact rescaling	What does the fixed original cutoff become after blow-up?	$\chi(F)$ becomes $\chi(X_0+\lambda^{-1}F_\lambda)$, with a harmless 4π factor	gives a rigorous corrected endpoint formula
literal scaled cutoff	Is $\phi_R(F_\lambda)$ the same fixed-cutoff object?	no; it is $\chi_{R,\lambda}(F)$ in original coordinates	exposes the missing uniform moving-family step
model obstruction	Is fixed compact scaled cutoff harmless?	no; flat stationary plane gives positive endpoint defect	disproves standalone lemma, but not full compact singular-flow case
annular flux audit	Can scaling make moving-cutoff errors small?	no; leading cutoff terms are order-one on a fixed scaled annulus	identifies the exact missing bridge
fixed-time compactness	Can earlier estimates give some compactness?	yes, fixed-time local Radon compactness	insufficient to compare two endpoint measures

5. Current status & next step

literal compact-scaled-cutoff version of equation (5.5) is not proved from the available pre-(5.5) material. The strongest unconditional corrected theorem is the fixed-original-cutoff version above. No full compact first-singular-flow counterexample has been established. The remaining actionable lemma is the following uniform moving-cutoff bridge.

Remaining lemma   uniform moving-cutoff bridge

Let $(\Sigma_t, F(\cdot,t))$ be a compact beta-symplectic critical surface flow smooth for $t < T$, let (X_0, T) be a singular spacetime point, fix $R > 0$, fix $\phi_R \in C_c^\infty(B_{2R}(0))$ with $\phi_R = 1$ on $B_R(0)$, and set

$$\chi_{R,\lambda}(X) = \phi_R(\lambda(X - X_0)).$$

For

$$Q_{R,\lambda}(t) = e^{c_1\sqrt{T-t}} \int_{\Sigma_t} (\cos\alpha)^{-p} \chi_{R,\lambda}(F) \rho_{X_0,T}(F,t)\, d\mu_t,$$

let

$$L_{R,\lambda} = \lim_{t \uparrow T} Q_{R,\lambda}(t),$$

assuming this limit exists from the one-sided estimate for each fixed λ. Prove, noncircularly and uniformly at the parabolic scale, that for every fixed $-1 < s_1 < s_2 < 0$,

$$Q_{R,\lambda}(T + \lambda^{-2}s_i) - L_{R,\lambda} \to 0 \quad (i = 1, 2).$$

Equivalently, prove the annular cutoff-flux terms supported on

$$R < |F_\lambda| < 2R$$

are $o(1)$, or cancel, as $\lambda \to \infty$.